
Notes for ‘Curve sketching’

Important Ideas and Useful Facts:

- (i) **Curve sketching, general principles:** *Curve sketching* involves drawing or sketching (often in one’s mind’s eye) the curve corresponding to the graph of a function $y = f(x)$ in the xy -plane. In the sketch,
- the scales on the x and y -axes do not need to be the same;
 - positioning of axes is flexible and chosen to highlight or enhance the main features of the curve;
 - aim for simplicity and clarity, and avoid including any unnecessary details;
 - focus on the main qualitative features;
 - extra details such as coordinates of important points can be added according to the particular problem one is trying to solve.
- (ii) **Curve sketching, checklist:** The following is a natural checklist of features to look for and include if possible when sketching the graph of a function $y = f(x)$:
- find the *y-intercept*, $f(0)$, where the curve crosses the y -axis, assuming $x = 0$ is in the domain of the function;
 - find any straightforward *x-intercepts*, where $f(x) = 0$, that is, where the curve crosses the x -axis (but be aware that typically this may not be easy, and indeed the purpose of creating a sketch may be to locate the approximate positions of x -intercepts, to assist in finding solutions or approximate solutions to a difficult equation);
 - look for *asymptotic behaviour*, seeing what happens as $x \rightarrow \infty$ or $x \rightarrow -\infty$, leading to possible horizontal or oblique asymptotes, and whether there are any vertical asymptotes, near which $y \rightarrow \infty$ or $y \rightarrow -\infty$;
 - look for where the curve is increasing ($y' > 0$) or decreasing ($y' < 0$), to help locate turning points and local or global maxima or minima, information for which is typically stored or summarised in the sign diagram for the derivative;
 - look for where the curve is concave up ($y'' > 0$) or concave down ($y'' < 0$) and points of inflection, where concavity changes, information for which is typically stored or summarised in the sign diagram for the second derivative.

Examples:

1. Sketch the curve $y = f(x) = 2x^3 - 9x^2 + 12x - 1$.

Solution: The y -intercept is $y(0) = -1$. (Finding the x -intercepts looks difficult, but we may obtain information about them as a consequence of the rest of this curve sketching exercise.)

The asymptotic behaviour is not very interesting. The cubic is dominated by the $2x^3$ term, so we have

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = -\infty ,$$

which are abbreviations for saying that $f(x)$ gets arbitrarily large and positive as x does, and $f(x)$ gets arbitrarily large (in magnitude) and negative as x does. In particular, there are no horizontal asymptotes.

You may be curious to know how we are sure there is no oblique asymptotic behaviour. Suppose, by way of contradiction that there exists some oblique asymptote with equation

$$y = mx + c$$

for some constants m and c . Certainly m can't be zero, for otherwise we would have a horizontal asymptote, and we have already seen that this is not the case. This hypothetical asymptote would have to have the property that the curve $y = f(x)$ gets arbitrarily close to it as x gets large, which would imply that the ratio of $mx + c$ with $f(x)$ should tend to 1. But, using our usual tricks for evaluating limits,

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{mx + c}{f(x)} &= \lim_{x \rightarrow \infty} \frac{mx + c}{2x^3 - 9x^2 + 12x - 1} = \lim_{x \rightarrow \infty} \frac{\frac{m}{x^2} + \frac{c}{x^3}}{2 - \frac{9}{x} + \frac{12}{x^2} - \frac{1}{x^3}} \\ &= \frac{0 + 0}{2 - 0 + 0 - 0} = 0 , \end{aligned}$$

which of course does not equal 1. A similar contradiction is reached by taking the limit as $x \rightarrow -\infty$, so there is no chance of there being an oblique asymptote.

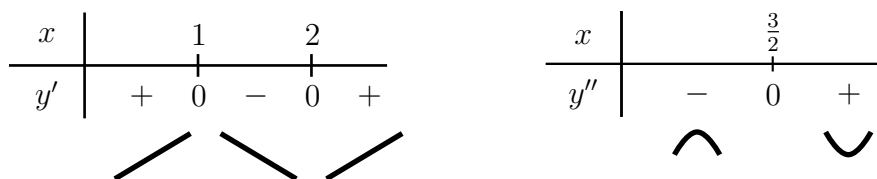
We have

$$y' = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) ,$$

so that $y' = 0$ when $x = 1, 2$. Further

$$y'' = 12x - 18 = 6(2x - 3) ,$$

so $y'' = 0$ when $x = \frac{3}{2}$. We get the following sign diagrams for the first and second derivatives:



The diagram for y' indicates that there is a turning point at $x = 1$, with local maximum

$$f(1) = 2 - 9 + 12 - 1 = 4 ,$$

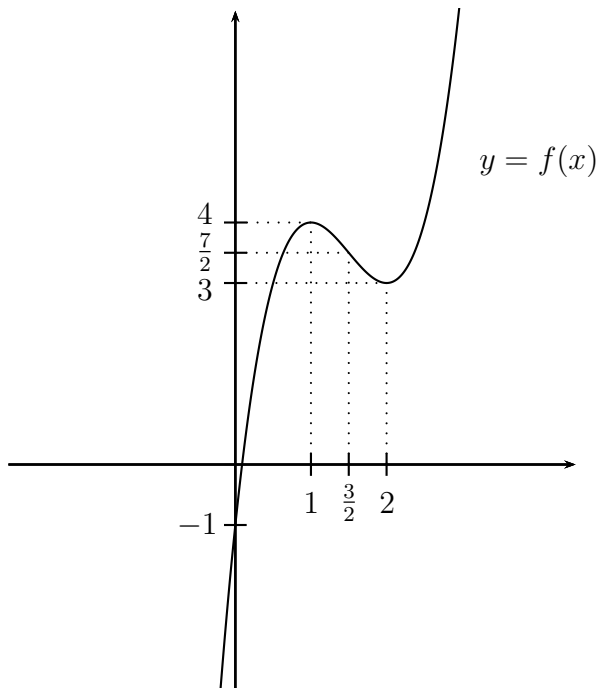
and a turning point at $x = 2$, with local minimum

$$f(2) = 16 - 36 + 24 - 1 = 3 .$$

The diagram for y'' indicates that there is a point of inflection at $x = \frac{3}{2}$, and the corresponding y -value is

$$f\left(\frac{3}{2}\right) = 2\left(\frac{3}{2}\right)^3 - 9\left(\frac{3}{2}\right)^2 + 12\left(\frac{3}{2}\right) - 1 = \frac{7}{2}.$$

All of these important features are highlighted in the sketch below. Notice that, in fact, there is only one x -intercept, something that was not at all evident from the rule of the function, and it is forced to lie somewhere between 0 and 1, and the sketch suggests quite close to 0.



- 2.** Sketch the graph of the rational function $y = g(x) = \frac{x^2 - 2}{x - 1}$.

Solution: The y -intercept is $g(0) = 2$. We have considered this curve before and exploited the fact that the rule can be expressed in different ways:

$$g(x) = \frac{x^2 - 2}{x - 1} = \frac{(x + \sqrt{2})(x - \sqrt{2})}{x - 1} = x + 1 - \frac{1}{x - 1} = x + 1 + \frac{1}{1 - x}.$$

The x -intercepts are clearly $\pm\sqrt{2}$. The line $x = 1$ is a vertical asymptote, and the line $y = x + 1$ is an oblique asymptote. Put $h(x) = \frac{1}{1-x}$. We have also seen before that the derivative is

$$g'(x) = 1 + h'(x) = 1 + \frac{1}{(1-x)^2}$$

with the following sign diagram:

$$\begin{array}{c|ccc} x & & 1 & \\ \hline g'(x) & + & u & + \end{array}$$

The derivative can also be put in the following form:

$$g'(x) = 1 + h'(x) = 1 + \frac{1}{(1-x)^2} = 1 + (1-x)^{-2} = 1 + (x-1)^{-2}.$$

The second derivative therefore becomes

$$g''(x) = \frac{d}{dx}(1) + \frac{d}{dx}((x-1)^{-2}) = \frac{d}{dx}((x-1)^{-2}).$$

We have been told that $\frac{d}{dx}(x^n) = nx^{n-1}$ for any exponent n , and it follows that

$$\frac{d}{dx}((x-1)^n) = n(x-1)^{n-1},$$

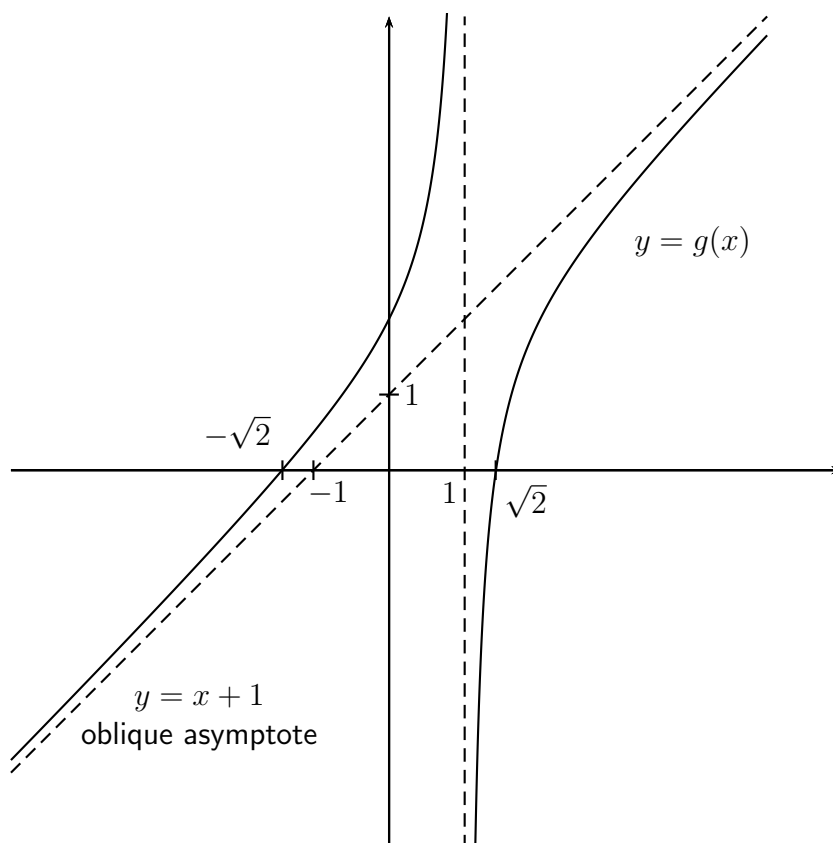
since the curve $y = (x-1)^n$ is obtained from the curve $y = x^n$ by a horizontal shift one unit to the right, so the slope of tangent lines is unaffected. We can apply this fact in our case, when $n = -2$, to conclude that

$$g''(x) = \frac{d}{dx}((x-1)^{-2}) = (-2)(x-1)^{-3} = \frac{-2}{(x-1)^3}.$$

It follows that $g''(1)$ is undefined and the sign of $g''(x)$ is the opposite of the sign of $x-1$, giving the following sign diagram:

x			1	
y''		$+$	u	$-$
		$\cup$		$\cap$

This pattern of concave up, followed by concave down, is matched by the concavity behaviour in the sketch below.



3. As a (difficult) variation of the previous example, sketch the graph of the rational function

$$y = f(x) = x + 1 + \frac{1}{x-1} = \frac{x^2}{x-1}.$$

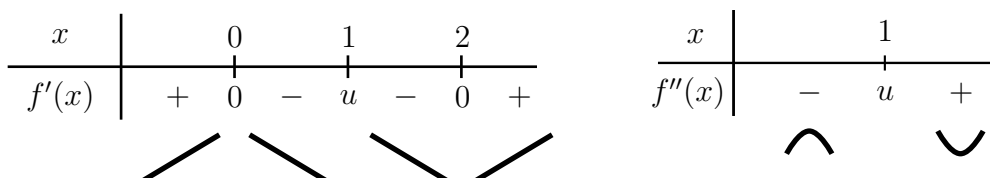
Solution: The y -intercept is $f(0) = 0$ and $y = 0$ when $x = 0$, so there is only one x -intercept and the curve passes through the origin. As in the previous example, the line $x = 1$ is a vertical asymptote and the line $y = x + 1$ is an oblique asymptote. The derivative becomes, similar to the previous example in the first three steps,

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x+1) + \frac{d}{dx}\left(\frac{1}{x-1}\right) = 1 + \frac{d}{dx}\left((x-1)^{-1}\right) = 1 - (x-1)^{-2} \\ &= 1 - \frac{1}{(x-1)^2} = \frac{(x-1)^2 - 1}{(x-1)^2} = \frac{x^2 - 2x + 1 - 1}{(x-1)^2} = \frac{x(x-2)}{(x-1)^2}. \end{aligned}$$

Differentiating again, we get

$$f''(x) = \frac{d}{dx}(1) - \frac{d}{dx}\left((x-1)^{-2}\right) = -(-2)(x-1)^{-3} = \frac{2}{(x-1)^3}.$$

From these formulae for $f'(x)$ and $f''(x)$ we get the following sign diagrams:



Thus there is a turning point at $x = 0$, with local maximum $y = 0$, and a turning point at $x = 2$, with local minimum $y = 4$. The curve is concave down for $x < 1$ and concave up for $x > 1$, and we get the following sketch:

